



Higher Diploma in Computer Science

Computer Systems & Networks

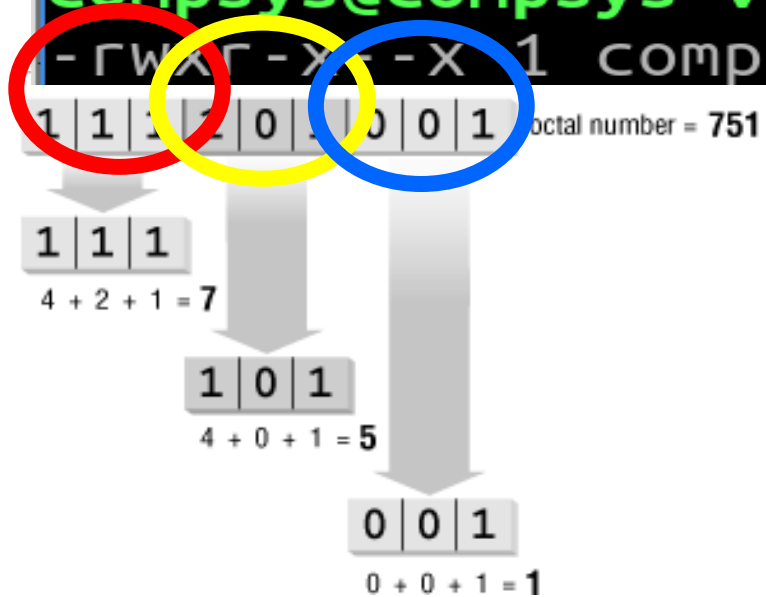


Octal Representation

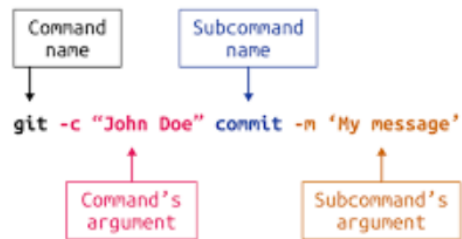
0	000	- - -	No permissions
1	001	- - x	Only Execute
2	010	- w -	Only Write
3	011	- w x	Write and Execute
4	100	r - -	Only Read
5	101	r - x	Read and Execute
6	110	r w -	Read and Write
7	111	r w x	Read, Write and Execute

Week 1: Octal Recap

```
compsys@compsys-virtualbox:~$ chmod 751 stats
compsys@compsys-virtualbox:~$ ls -l st*
-rwxr-xr-x 1 compsys compsys 248 Aug 29 13:54 stats
```



01: Command Line



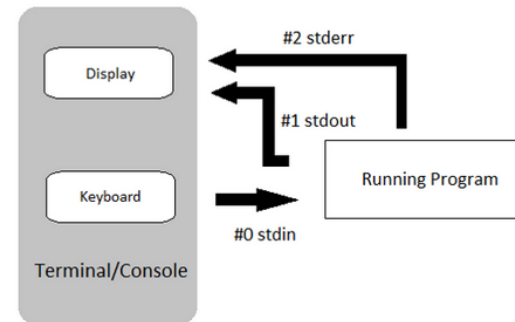
Commands · Arguments ·
Options · Variables

02: File Management



Globs · Copy · Move ·
Remove

03: IO Redirection



STDIN · STDOUT · STDERR

Lab



File Management · IO
Redirection · Search &
Filter text



Command Line Skills

Introduction

- This class will cover the basics of the command line such as:
 - The command line interface
 - The shell
 - Commands
 - Variables
 - Command Types
 - Quoting
 - Control Statements





Command Line Interface

Command Line Interface

The Linux community promotes the CLI due to its power, speed and ability to accomplish a vast array of tasks with a **single command line** instruction.

The CLI provides

- more precise control,
- greater speed and
- the ability to automate tasks more easily through scripting.



The Shell

The Shell

- Once a user has entered a command the terminal then accepts what the user has typed and passes to a *shell*.
- The shell is the CLI that translates commands entered by a user into actions to be performed by the OS.

```
#!/bin/bash
# This is a description of my script...

date
who | wc -l
pwd
```

The Shell

The Bash shell also has many popular features, such as:

- Command line history
- Inline editing
- Scripting
 - The ability to place commands in a file and then interpret (effectively use Bash to execute the contents of) the file, resulting in all of the commands being executed.
- Aliases
 - The ability to create short nicknames for longer commands.
- Variables
 - Used to store information for the Bash shell and for the user.

The Shell

Typically the prompt contains information about the user and the system. Below is a common prompt structure:



```
compsys@compsys-virtualbox:~
```

A typical prompt shown contains the following information:

Username (sysadmin)	System name (localhost)	Current Directory (~)
------------------------	----------------------------	--------------------------



Commands

Commands

```
compsys@compsys-virtualbox:~$ ls  
Desktop Documents Downloads Music Pictures Public Templates Videos
```

A command is a software program that when executed on the CLI, performs an action on the computer.

To execute a command, the first step is to type the name of the command.

If you type `ls` and hit **Enter**. The result should resemble the example below:

Commands

```
command [options] [arguments]
```

Some commands require additional input to run correctly.

This additional input comes in two forms: *options* and *arguments*.

- **Options** are used to modify the core behavior of a command.
- **Arguments** are used to provide additional information (such as a filename or a username).



Options

Options

```
command [options] [arguments]
```

used with commands to expand or modify the way a command behaves.

```
compsys@compsys-virtualbox:~$ ls -l
```

```
total 40
```

```
drwxr-xr-x 2 compsys compsys 4096 Sept 29  20:21 Desktop
```

```
drwxr-xr-x 1 compsys compsys 4096 Sept 29  20:25 Documents
```

```
compsys@compsys-virtualbox:~$ ls -lr
```



Arguments

Arguments

```
command [options] [arguments]
```

... used to specify something for the command to act upon.

If the `ls` command is given the name of a directory as an argument, it lists the contents of that directory

Some commands (such as `ls`) accept **multiple** arguments

```
compsys@compsys-virtualbox:~$ ls /etc/ppp  
ip-down.d  ip-up.d
```

```
compsys@compsys-virtualbox:~$ ls /etc/ppp /etc/ssh
```




Variables

Variables

A variable is a feature that allows the user or the shell to store data.



Variables are given names and stored temporarily in memory.



There are two types of variables used in the Bash shell,

local and

environment.

Local Variables

- Local or *shell*, variables **exist only in the current shell**.
 - When the user closes a terminal window or shell, all of the variables are lost.
 - To set the value of a variable, use the following assignment expression.

```
variable=value
```

- Create a local variable and assign a value:

```
compsys@compsys-virtualbox:~$ variable1='HDip are ready'
```

- Display the value of the variable using a dollar sign \$ character followed by the variable name as an argument to the `echo` command:

```
compsys@compsys-virtualbox:~$ echo $variable1
HDip are ready
```

Environment Variables

- *Environment variables*, also called *global variables*, are available system-wide.
- Examples include the `PATH`, `HOME`, and `HISTSIZE` variables.

```
compsys@compsys-virtualbox:~$ echo $HISTSIZE
```

```
1000
```

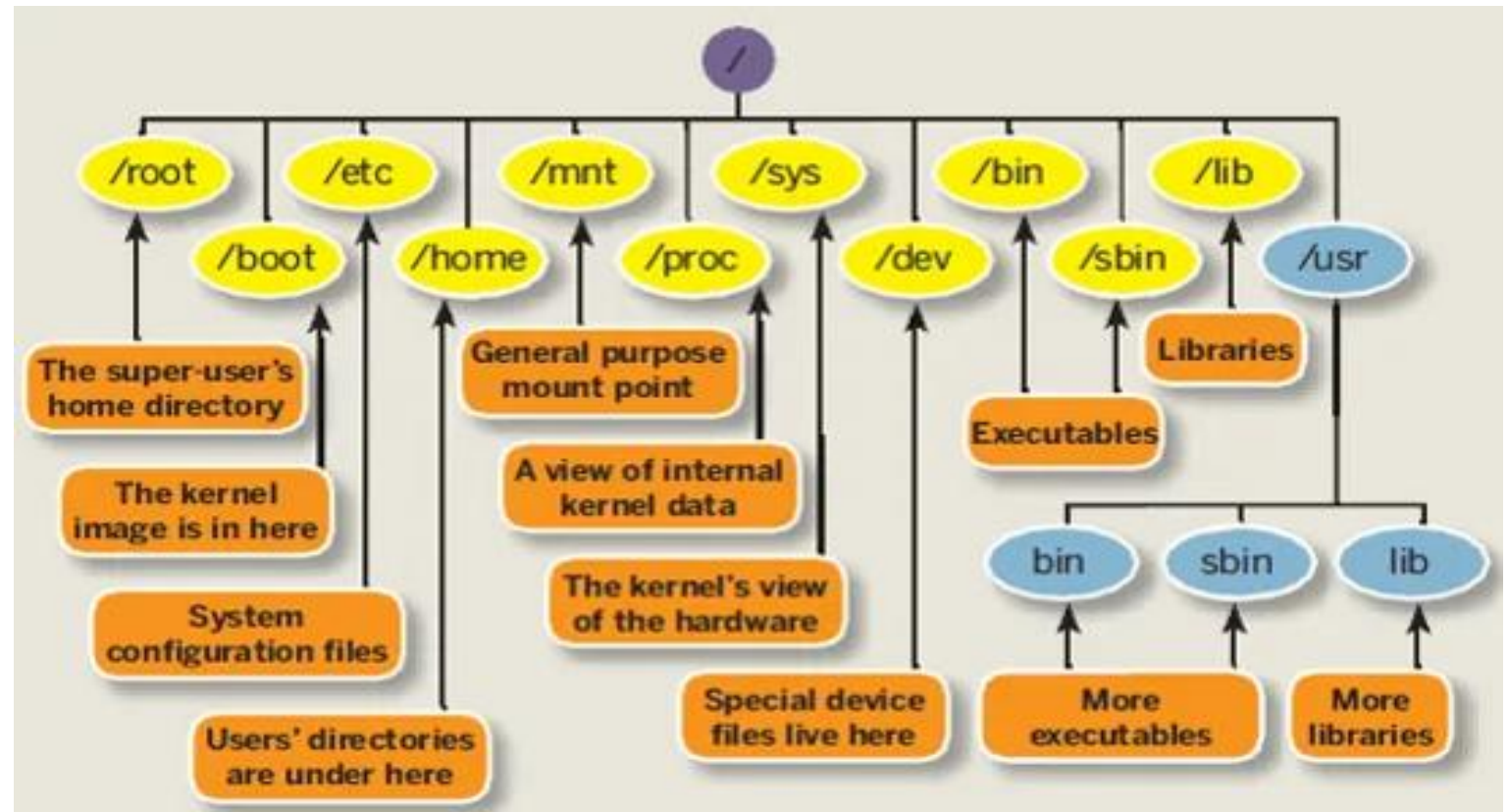
- The `env` command outputs a list of the environment variables

Path Variable

- A very important Bash shell variables to understand is the `PATH` variable.
 - The `PATH` variable lists all the places that the system can look for programs to execute.
 - EXERCISE: Display the path of the current shell:

```
compsys@compsys-virtualbox:~$ echo $PATH
/home/compsys/.config/nvm/versions/node/v14.3.0/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
```

Filesystem Hierarchy Structure (FHS)

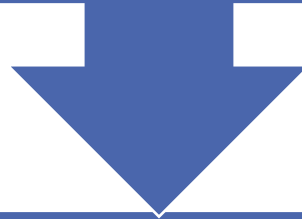




Command Types

Command Types

The type command can be used to determine information about command type.



There are several different sources of commands within the shell of your CLI:

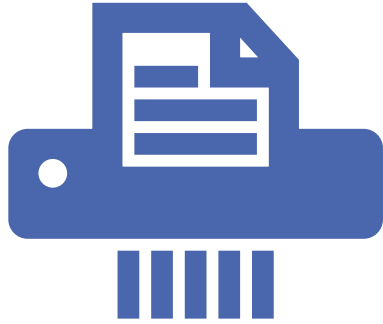
Internal commands	External commands	Aliases	Functions
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Internal (built-in) Commands

- Also called *built-in* commands, these commands are built into the shell itself.
 - `cd` (change directory) command as it is part of the Bash shell.
- The `type` command identifies the `cd` command as an internal command:

```
compsys@compsys-virtualbox:~$ type echo
echo is a shell builtin
compsys@compsys-virtualbox:~$ type cat
cat is /usr/bin/cat
```

External Commands



External commands are stored in files that are searched by the shell.



The `which` command searches for the **location** of a command by searching the PATH variable.

```
compsys@compsys-virtualbox:~$ which ls  
/usr/bin/ls
```

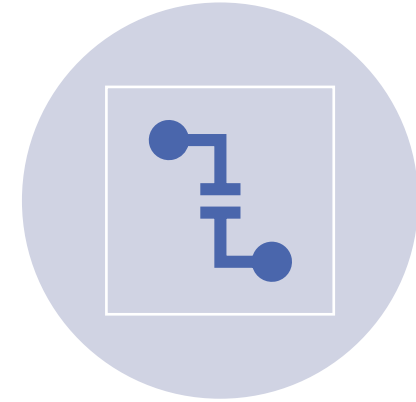
External Commands



EXTERNAL COMMANDS CAN BE EXECUTED
BY TYPING THE COMPLETE PATH TO THE
COMMAND.



FOR EXTERNAL COMMANDS, THE TYPE
COMMAND DISPLAYS THE LOCATION OF
THE COMMAND:



TO DISPLAY ALL LOCATIONS THAT CONTAIN
THE COMMAND NAME, USE THE -A
OPTION TO THE TYPE COMMAND:

```
compsys@compsys-virtualbox:~$ /bin/ls
```

```
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
```

```
compsys@compsys-virtualbox:~$ type mkdir
mkdir is /usr/bin/mkdir
```

```
compsys@compsys-virtualbox:~$ type -a echo
```

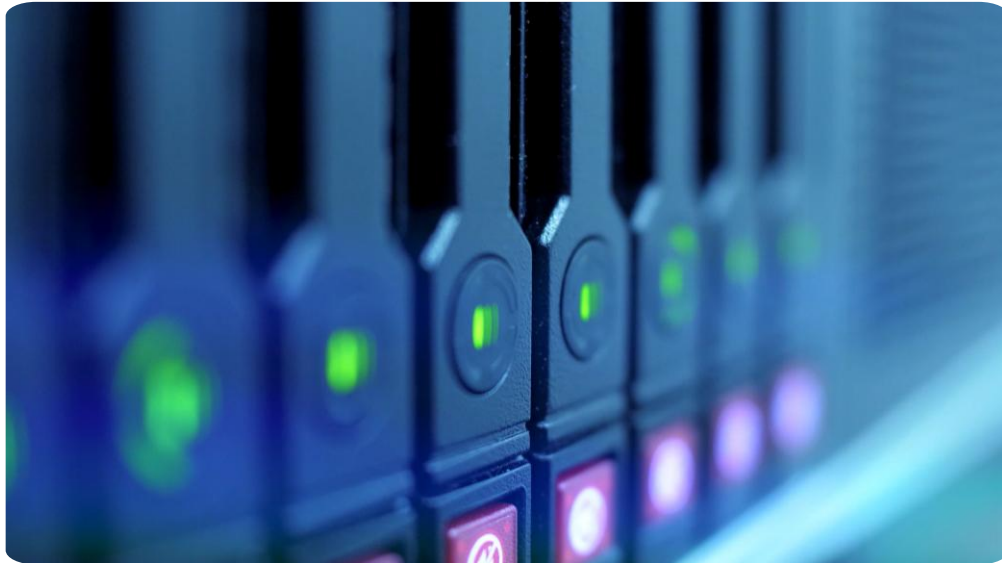
Aliases

- An *alias* can be used to map longer commands to shorter key sequences.
- To determine what aliases are set on the current shell use the `alias` command:
- The `type` command can identify aliases to other commands:

```
compsys@compsys-virtualbox:~$ alias
alias alert='notify-send --urgency=low -i "${[ $? = 0 ] &&
)" "$(history|tail -n1|sed -e '\''s/^\s*[0-9]\+\s*//;s/[:;8
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -alF'
alias ls='ls --color=auto'
```

```
compsys@compsys-virtualbox:~$ type ll
ll is aliased to `ls -alF'
```

Functions



- Functions can also be built using existing commands to:
 - Create new commands
 - Override commands built-in to the shell or commands stored in files